MILITARY SPECIFICATION

SABOT, 81MM PRACTICE: M1 ASSEMBLING AND PACKING

This specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the manufacture of parts, assembly and packing of one type of training round for the 81MM Mortar System.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitations, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables
MIL-STD-810	-	Environmental Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of the specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

DRAWINGS

9276596	-	Container Ammo Fiber M252A5 for 81MM Mortar Ammunition
9311182	-	Sabot, 81MM, Practice: M1
12551873	-	Packing and Marking for Container, Ammunition, Fiber for Sabot, 81MM, Practice: M1
12556165	-	Box, Wood, Packing for Container, Ammunition, Fiber for Sabot, 81MM, Practice: M1
12556166	-	Packing and Marking for Box, Wood, for Container, Ammunition Fiber for Sabot, 81MM, Practice: M1

(Copies of specifications, standards, drawings, and publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of conflict between the text of this specification and the reference cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supercede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Assembly. The sabot assembly shall comply with all requirements specified on drawing (dwg.) 9311182 and associated drawings and with all requirements specified in applicable specifications and standards.

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3.3 Ballistic test. The Flange Nut Assembly shall function the subcaliber cartridge and shall be reuseable when test fired in the applicable equipment and subjected to temperature conditions of $0^{o}F$ and plus $110^{o}F$.

3.4 Humidity test. The Sabot assembly shall withstand exposure to a humid atmosphere when tested per MIL-STD-810, procedure V (limited to three cycles).

3.5 Five foot drop test. The Flange Nut Assembly shall not function the subcaliber cartridge when dropped nose down from a height of five feet when assembled to an M1 Sabot and a Jet Body (dwg. 9299084) with Jet Nozzle assembly (dwg. 9299088). The Flange Nut Assembly shall be reuseable after being dropped.

3.6 Workmanship. The item of shipment, its components, and materials shall be free from dirt, chips, grease and other foreign matter. There shall be no cracks, scratches, dents, gouges, holes, porous areas, distortion, or undesirable characteristics which would in any way detract from the useability of the product for its intended purpose. Parts and assemblies shall be free from all contaminants and damage after cleaning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set for in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection (see 4.3)

b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the components in the quantities indicated in Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE: Sabot, 81MM Practice, M1		SHEET 1 OF 3		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Sabot, 81MM, Practice, M1 (Dwg. 9311182) Examination for defects	3		3.2(a)	4.4.2.1/4.4.2.1.1
	Humidity Test	3		3.4	4.5.4
	Body (Dwg. 9311183) Examination for defects	5		3.2	4.4.2.2
	Tail Fin (Dwg. 9311188) Examination for defects	5		3.2	4.4.2.3
	Nose (Dwg. 9311187) Examination for defects	5		3.2	4.4.2.4
	Segment (Dwg. 9311184) Examination for defects	5		3.2	4.4.2.5
	Body and Segment Assembly (Dwg. 9311190) Examination for defects	5		3.2	4.4.2.6
	Flange Nut (Dwg. 9369936) Examination for defects	10		3.2	4.4.2.7

NOTES:

(a) Use 4.4.2.1 or 4.4.2.1.1 as applicable.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
	Sabot, 81MM Practice, M1 (Cont)				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Barrel No. 1 (Dwg. 9311185)				
	Examination for defects	5		3.2	4.4.2.8
	Barrel No. 2 (Dwg. 9311186)				
	Examination for defects	5		3.2	4.4.2.9
	Barrel (one piece) (Dwg. 9311196)				
	Examination for defects	5		3.2	4.4.2.10
	Pin Firing (Dwg. 12630564)				
	Examination for defects	10		3.2	4.4.2.11
	Flange Nut Assembly (Dwg. 9311189)				
	Examination for defects	10		3.2	4.4.2.12
	Ballistic test	15		3.3	4.5.2
	Five foot drop test	15(b)		3.5	4.5.3
	Housing (Dwg. 9369935)				
	Examination for defects	10		3.2	4.4.2.13

NOTES: (b) The Five Foot Drop Test shall be run after the Ballistic Test on the same 15 unit samples. The same sample shall be conditioned to the required temperature and tested per the above. (c) The spring embrittlement test shall be performed after the examination for defects and spring rate test.

AMSMC Form 1570a, 1 Apr 85

Replaces DRDAR-QA Form 160a, 1 Jun 83, which may not be used.

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-S-63110A (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
	Sabot, 81MM Practice, M1 (Cont.)				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Spring (Dwg. 9369938)				
	Examination for defects	30		3.2	4.4.2.14
	Spring rate test	30		3.2	4.5.5
	Embrittlement test	30(c)		3.2	4.5.6
	Nut (Dwg. 12630566)				
	Examination for defects	10		3.2	4.4.2.15
	Sleeve, locking (Dwg. 12630565)				
	Examination for defects	10		3.2	4.4.2.16
	Firing pin assembly (Dwg. 12630563)				
	Examination for defects	10		3.2	4.4.2.17

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Sabot, 81MM, Practice, M1 (With Double Barrel)				9311182
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside profile and alignment of barrels		100%	3.2	Gage
102	Inside diameter for entire length		100%	3.2	Plug Gage
103	Barrel No. 1 fully seated into barrel No. 2 (prior to assembly to sabot)		100%	3.2	Visual
104	Tail fin torque		0.40%	3.2	Gage
105	Nose torque		0.40%	3.2	Gage
106	Setscrew torque		0.40%	3.2	Gage
107	Rear barrel threads damaged		0.40%	3.2	Visual
108	Segments missing or loose		100%	3.2	Visual/Manual
109	Presence of adhesive on inside diameter		100%	3.2	Visual
110	Overall length of sabot		0.40%	3.2	Gage
111	Runout of forward to rear inside diameter		0.40%	3.2	Gage
112	Profile and alignment of sabot with barrels		100%	3.2	Gage
113	Improper alignment of body and fin		100%	3.2	Chamber Gage
Minor					
201	Presence of nicks or gouges on interior or exterior		0.65%	3.2	Visual
202	Poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1.1	Sabot, 60MM, Practice, M1 (with single barrel)	1 OF 1	9311182
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside profile and alignment of barrels		100%	3.2	Gage
102	Inside diameter for entire length		100%	3.2	Plug Gage
103	Overall length		0.40%	3.2	Gage
104	Tail fin torque		0.40%	3.2	Gage
105	Nose torque		0.40%	3.2	Gage
106	Setscrew torque		0.40%	3.2	Gage
107	Rear barrel threads damaged		0.40%	3.2	Visual
108	Segments missing or loose		100%	3.2	Visual/Manual
109	Profile and alignment of sabot with barrel		100%	3.2	Gage
110	Improper alignment of body and fin		100%	3.2	Chamber Gage
Minor					
201	Presence of nicks or gouges on interior or exterior surface		0.65%	3.2	Visual
202	Poor workmanship		0.65%	3.6	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311183
4.4.2.2	Body				NEXT HIGHER ASSEMBLY 9311190
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter, max.		0.40%	3.2	Gage
102	Forward inside diameter		0.40%	3.2	Gage
103	Rear inside diameter		0.40%	3.2	Gage
104	Runout of rear inside diameter with forward inside diameter		0.40%	3.2	Gage
105	Width of gas checks (4 places)		0.40%	3.2	Gage
106	Depth of gas checks (4 places)		0.40%	3.2	Gage
107	Width of slots (4 places)		0.40%	3.2	Gage
108	Depth of slots (4 places)		0.40%	3.2	Gage
109	Radius of slots (4 places)		0.40%	3.2	Gage
110	Overall length		0.40%	3.2	Gage
111	Perpendicularity of forward inside diameter to front face		0.40%	3.2	Gage
112	Inside central diameter		0.40%	3.2	Gage
Minor					
201	Presence of burrs		0.65%	3.6	Visual
202	Surface finish improper		0.65%	3.6	Visual/Manual
203	Poor workmanship		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Tail Fin	1 OF 1	9311188
			NEXT HIGHER ASSEMBLY: 9311182

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Forward inside diameter		0.40%	3.2	Gage
102	Set screw threads missing or damaged		0.40%	3.2	Visual
103	Diameter of fins		0.40%	3.2	Gage
104	Pitch diameter of rear threads		0.40%	3.2	Gage
105	Minor diameter of rear threads		0.40%	3.2	Gage
106	Symmetry of fins to forward inside diameter and pitch diameter of rear threads		0.40%	3.2	Gage
107	True position of fins to forward inside diameter		0.40%	3.2	Gage
108	Perpendicularity of forward face with forward inside diameter		0.40%	3.2	Gage
Minor					
201	Surface finish improper		0.65%	3.2	Visual/Manual
202	Poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Nose				9311187
					NEXT HIGHER ASSEMBLY
					9311182
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Rear inside diameter		0.40%	3.2	Gage
102	Threads missing or damaged		0.40%	3.2	Visual
103	Pitch diameter of threads		0.40%	3.2	Gage
104	Minor diameter of threads		0.40%	3.2	Gage
105	Runout of rear inside diameter with the pitch diameter of the threads		0.40%	3.2	Gage
106	Overall length		0.40%	3.2	Gage
Minor					
201	Presence of burrs		0.65%	3.6	Visual
202	Protective finish missing or inadequate		0.65%	3.1	Visual
203	Cut-out flat missing		0.65%	3.2	Visual
204	Poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Segment		9311184
			NEXT HIGHER ASSEMBLY: 9311190

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length		0.40%	3.2	Gage
102	Width		0.40%	3.2	Gage
103	Thickness		0.40%	3.2	Gage
Minor					
201	Presence of burrs		0.65%	3.6	Visual
202	Poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Body and Segment Assembly				9311190
					NEXT HIGHER ASSEMBLY
					9311182
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Runout of segments with forward inside diameter and rear inside diameter		0.40%	3.2	Gage
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Flange Nut				9369936
					NEXT HIGHER ASSEMBLY 9311189
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of threads		0.40%	3.2	Gage
102	Minor diameter of threads		0.40%	3.2	Gage
103	Diameter of thread relief		0.40%	3.2	Gage
104	Small rear inside diameter		0.40%	3.2	Gage
105	Depth of thread shoulder		0.40%	3.2	Gage
106	Runout of the pitch diameter of the thread with the rear inside diameter		0.40%	3.2	Gage
107	Knurl missing		0.40%	3.2	Gage
108	Overall length		0.40%	3.2	Gage
109	Large rear inside diameter		0.40%	3.2	Gage
110	Depth of rear inside diameter		0.40%	3.2	Gage
111	Small forward inside diameter		0.40%	3.2	Gage
112	Depth of forward inside diameter		0.40%	3.2	Gage
113	Depth of forward thread relief		0.40%	3.2	Gage
114	Diameter of forward thread relief		0.40%	3.2	Gage
Minor					
201	Protective finish missing or inadequate		0.65%	3.1	Visual
202	Inside diameter of aft cavity		0.65%	3.2	Gage
203	Poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Barrel No. 1	1 OF 1	9311185
			NEXT HIGHER ASSEMBLY: 9311182

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of forward thread		0.40%	3.2	Gage
102	Major diameter of forward thread		0.40%	3.2	Gage
103	Pitch diameter of rear thread		0.40%	3.2	Gage
104	Major diameter of rear thread		0.40%	3.2	Gage
105	Outside diameter		0.40%	3.2	Gage
106	Runout of the pitch diameter of the forward thread with the outside diameter		0.40%	3.2	Gage
107	Runout of the pitch diameter of the rear thread with the outside diameter		0.40%	3.2	Gage
108	Inside diameter excluding stepped area		0.40%	3.2	Gage
109	Runout of inside diameter with outside diameter		0.40%	3.2	Gage
110	Overall length		0.40%	3.2	Gage
111	Rear inside diameter		0.40%	3.2	Gage
112	Distance from aft face to forward face of center inside step		0.40%	3.2	Gage
Minor					
201	Presence of burrs		0.65%	3.6	Visual
202	Protective finish missing		0.65%	3.2	Visual
203	Chamfered grooves missing		0.65%	3.2	Visual
204	Poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Barrel No. 2	1 OF 2	9311186
			NEXT HIGHER ASSEMBLY: 9311182

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Forward outside diameter		0.40%	3.2	Gage
102	Center inside diameter		0.40%	3.2	Gage
103	Outside diameter forward of set screw groove (1.510 dimension)		0.40%	3.2	Gage
104	Runout of central inside diameter with forward outside diameter and outside diameter forward of set screw groove		0.40%	3.2	Gage
105	Runout of forward outside diameter with outside diameter forward of set screw groove		0.40%	3.2	Gage
106	Rear inside diameter		0.40%	3.2	Gage
107	Runout of rear inside diameter with center inside diameter		0.40%	3.2	Gage
108	Pitch diameter of rear thread		0.40%	3.2	Gage
109	Major diameter of rear thread		0.40%	3.2	Gage
110	Runout of the pitch diameter of the rear thread with the outside diameter forward of the set screw groove and the forward outside diameter		0.40%	3.2	Gage
111	Overall length		0.40%	3.2	Gage
112	Setscrew groove missing		0.40%	3.2	Visual
113	Pitch diameter of forward inside thread		0.40%	3.2	Gage
114	Minor diameter of forward inside thread		0.40%	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.9	Barrel No. 2 (Cont)				9311186
					NEXT HIGHER ASSEMBLY
					9311182
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
115	Runout of pitch diameter of forward thread with forward outside diameter & undercut of forward thread		0.40%	3.2	Gage
Minor					
201	Presence of burrs		0.65%	3.6	Visual
202	Protective finish missing		0.65%	3.2	Visual
203	Poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.10	Barrel (one piece)				9311196
					NEXT HIGHER ASSEMBLY 9311182
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of rear thread		0.40%	3.2	Gage
102	Major diameter of rear thread		0.40%	3.2	Gage
103	Pitch diameter of forward thread		0.40%	3.2	Gage
104	Major diameter of forward thread		0.40%	3.2	Gage
105	Forward outside diameter		0.40%	3.2	Gage
106	Central outside diameter (1.457 dimension)		0.40%	3.2	Gage
107	Runout of central outside diameter with forward outside diameter		0.40%	3.2	Gage
108	Outside diameter forward of set screw groove (1.510 dimension)		0.40%	3.2	Gage
109	Runout of 1.510 dimension with forward outside diameter		0.40%	3.2	Gage
110	Runout of front thread with forward outside diameter		0.40%	3.2	Gage
111	Runout of rear thread with forward outside diameter and 1.510 dimension		0.40%	3.2	Gage
112	Forward inside diameter		0.40%	3.2	Gage
113	Runout of inside diameter with outside diameter and 1.510 dimension		0.40%	3.2	Gage
114	Runout of forward inside diameter with center inside diameter		0.40%	3.2	Gage
115	Center inside diameter (.9465 dimension)		0.40%	3.2	Gage

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.10	Barrel (one piece) (Cont)				9311196
					NEXT HIGHER ASSEMBLY
					0311182
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
116	Runout of center inside diameter with forward outside diameter and 1.510 dimension		0.40%	3.2	Gage
117	Rear inside diameter		0.40%	3.2	Gage
118	Runout of rear inside diameter with 0.9465 dimension		0.40%	3.2	Gage
119	Length from aft face to aft end of last inside central step		0.40%	3.2	Gage
120	Length from aft face to forward end of middle inside central step		0.40%	3.2	Gage
121	Inside surface finish improper		0.40%	3.2	Gage
Minor					
201	Protective finish missing		0.65%	3.1	Visual
202	Overall length		0.65%	3.2	Gage
203	Poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Pin, Firing	1 OF 1	12630564
			NEXT HIGHER ASSEMBLY 12630563

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length from aft end to aft of head		0.40%	3.2	Gage
102	Diameter of firing pin shaft		0.40%	3.2	Gage
103	Major diameter of aft shaft thread		0.40%	3.2	Gage
104	Pitch diameter of aft shaft thread		0.40%	3.2	Gage
105	Length of aft shaft thread, min		0.40%	3.2	Gage
106	Runout of thread and firing pin shaft		0.40%	3.2	Gage
107	Hardness of firing pin	25(a)		3.2	Gage
Minor					
201	Improper profile of head		0.65%	3.2	Gage
202	Improper surface finish		0.65%	3.2	Visual
203	Poor workmanship		0.65%	3.6	Visual

NOTES:
(a) From each heat treatment batch.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Flange Nut Assembly				9311189
					NEXT HIGHER ASSEMBLY 9311182
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Firing pin does not move freely when depressed from rear		100%	3.2	Manual
102	Parts missing		100%	3.2	Visual
Minor					
201	Improper application of adhesive		0.65%	3.2	Visual
202	Poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	*Housing*				9369935
					NEXT HIGHER ASSEMBLY 9311189
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter aft of head		0.40%	3.2	Gage
102	Length from aft end to head		0.40%	3.2	Gage
103	Length of head		0.40%	3.2	Gage
104	Diameter of head		0.40%	3.2	Gage
105	Diameter of hole thru head		0.40%	3.2	Gage
106	Improper angle of head face		0.40%	3.2	Gage
107	Diameter of aft shaft cavity		0.40%	3.2	Gage
108	Depth of aft shaft cavity		0.40%	3.2	Gage
109	Improper aft shaft thread		0.40%	3.2	Gage
110	Length of aft shaft thread		0.40%	3.2	Gage
111	Improper interior finish		0.40%	3.2	Visual
112	Runout of aft shaft cavity with diameter aft of head		0.40%	3.2	Gage
113	Runout of hole thru head with diameter aft of head		0.40%	3.2	Gage
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.14	Spring		9369938
			NEXT HIGHER ASSEMBLY
			12630563

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Spring rate test	25(a)	0.40%	3.2	Test Para 4.5.5
102	Spring embrittlement test			3.2	Test Para 4.5.6
103	Spring diameter over max		0.40%	3.2	Gage
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES:

(a) Each known heat of springs shall conform to this test. Any spring failure shall constitute a failure of that spring heat lot.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Nut				12630566
					NEXT HIGHER ASSEMBLY
					12630563
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Overall length		0.40%	3.2	Gage
103	Improper external cylindrical surface finish		0.40%	3.2	Gage
104	Internal thread improper		0.40%	3.2	Gage
105	Runout of threads to outside diameter		0.40%	3.2	Gage
106	Thread improper		0.40%	3.2	Gage
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.16	Sleeve, Locking				12630565
					NEXT HIGHER ASSEMBLY 9311189
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		0.40%	3.2	Gage
102	Outside diameter		0.40%	3.2	Gage
103	Location of wrench slots		0.40%	3.2	Gage
104	Improper internal thread		0.40%	3.2	Gage
Minor					
201	Depth of wrench slots		0.65%	3.2	Gage
202	Width of wrench slots		0.65%	3.2	Gage
203	Poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63110A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630563
4.4.2.17	Firing Pin Assembly				NEXT HIGHER ASSEMBLY 9311189
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Firing pin does not move freely when depressed from rear		100%	3.2	Manual
102	Parts missing		100%	3.2	Visual
Minor					
201	Improper application of adhesive		0.65%	3.2	Visual
202	Poor workmanship		0.65%	3.6	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

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4.4.3 Testing. See 4.5.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspections and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Ballistic test. The test area shall not be rocky or hard-packed ground surface. The impact area shall use a tarpaulin or canvas covering with the required size and placed centrally two meters from the mortar tube. The flange nut assembly shall be assembled with a production accepted sabot and 22mm subcaliber cartridge, conditioned a minimum of four (4) hours at the required temperature and test fired within two (2) minutes after removal from the conditioning chamber. All units shall be fired at an elevation of 800 mils. Any Government accepted 22mm subcaliber cartridge (any charge) is acceptable for this test. If one or more rounds fail to fire due to flange nut assembly failures, the First Article sample shall be rejected.

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4.5.3 Five foot drop test. This test shall be run subsequent to the ballistic test in paragraph 4.5.2. The flange nut assembly shall be assembled with a production accepted sabot and a Jet Body (dwg. 9299084) with Jet Nozzle assembly (dwg. 9299088) (Government Furnished Material (GFM). The round shall be conditioned a minimum of four (4) hours at the required temperature and dropped within two (2) minutes after removal from the conditioning chamber. The assembled unit shall then be suspended nose down over a smooth, uniform concrete surface (three (3) feet on a side minimum and one (1) foot thick minimum). The round shall be released by a quick-release mechanism onto the surface. If one or more rounds fire or the primer is dented due to flange nut assembly failure, the First Article sample shall be rejected.

4.5.4 Humidity test. The Sabot assembly (inert loaded) shall be subjected to the required test. Observation shall be made as to whether the flange nut can be readily disassembled and the Sabot can be properly assembled. This test is for information only.

4.5.5 Spring rate test. Each spring shall be compressed to a height of .4375 ± .01 inch. The compressed load gage reading shall not exceed 5.5 ± 0.05 pounds. Any spring that fails this requirement shall be classified defective. This test shall be performed at ambient temperature. This is not a destructive test.

4.5.6 Spring embrittlement test. Each spring shall be stretched along its longitudinal axis until the spring no longer returns to its original position. When the spring no longer returns to its original length or profile, it shall be considered distorted permanently. Any spring that breaks prior to this point shall be classified defective. One defective spring shall constitute rejection of that particular spring heat lot. This is a destructive test.

5. PACKAGING

5.1 Packaging and marking requirements. Packaging and marking shall be as specified on dwgs. 12551873 and 9276596.

5.2 Packing and marking requirements. Packing and marking shall be as specified on dwgs. 12556165 and 12556166.

6. NOTES

6.1 Intended use. The Sabots covered by this specification are intended for use on the 81MM Mortar Training Device.

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6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I, Dover, NJ 07801-5001.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now uner the cognizance of ARDEC.

6.5 22MM subcaliber cartridges. The procuring agency will provide the required 22mm subcaliber cartridges for the Ballistic Test.

6.6 Jet body with jet nozzle assembly. The procuring agency will provide the required jet body (dwg. 9299084) with jet nozzle assembly (dwg. 9299088) for the Five Foot Drop Test.

6.7. Subject term (key word) listing.

Superseding
Sabot
AMSC number
Military specification
Requirements
Revision
Shipment
Inspection
Compliance
Conformance

6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-A805)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-63110A (AR)	

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

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